

Day 39 - DHCP

Sunday, September 6, 2026

10:46 AM

DHCP Info

DHCP General Information

Dynamic Host Configuration Protocol (DHCP) is a protocol that allows hosts to automatically/dynamically learn various aspects of their network configuration without manual configuration, including information such as:

- *IP address*
- *Subnet mask*
- *Default gateway*
- **DNS server**
- **DHCP** is typically used for '*client devices*', while devices like routers, servers and etc, are manually configured
- **DHCP** servers run on port **67** while **DHCP** clients run on port **68**.

What devices are usually DHCP servers?

- In small networks (like homes and small offices), the router usually acts as the **DHCP** server.
- In large networks, a dedicated server usually running Windows/Linux acts as the **DHCP** server.

DHCP *server leasing*

DHCP servers '*lease*' IP addresses to clients.

This means that any IP addresses assigned to clients are not permanent, and the client must give up the address at the end of the lease.

DHCP DORA Steps

For a DHCP client to contact and get information from the DHCP server, it has to go through four steps sometimes called '*DORA*' for short:

1. **DHCP Discover**
2. **DHCP Offer**
3. **DHCP Request**
4. **DHCP Ack**

1. DHCP Discover

A DHCP *Discover* is the first message that clients send when they are looking for a DHCP server to connect to. Some of the message properties are:

- It is a **broadcast** message
- The *source MAC* is the client's MAC address
- The *destination MAC* is the **broadcast** MAC address (*ffff.ffff.ffff*)
- The *source IP* is **0.0.0.0**
- The *destination IP* is the **broadcast** IP address (**255.255.255.255**)

The message contains some general information about the client for the DHCP *server* to use, like whether the client wants the response messages sent back to be unicast or broadcast, or if the client would like a specific IP address.

2. DHCP Offer

The DHCP *Offer* message is sent by the DHCP server after receiving a DHCP *Discover* message. It offers an IP address to the client. Some of the message properties are:

- It can be a **broadcast** or a **unicast** message.
- The *source MAC* is the server's MAC address
- The *destination MAC* is the client's MAC address, which the server learned from the *Discover* message.
- The *source IP* is the IP address of the server
- The *destination IP* is the IP address that's being offered to the client

3. DHCP Request

The DHCP *Request* is a message that is sent from the client to a DHCP server, informing the server that it is requesting the IP address that has been previously offered to it. Some of the message properties are:

the IP address that has been previously offered to it. Some of the message properties are:

- It is a **broadcast** message
- The **source MAC** is the client's MAC address
- The **destination MAC** is the **broadcast** MAC address (**ffff.ffff.ffff**)
- The **source IP** is **0.0.0.0**
- The **destination IP** is the **broadcast** IP address (**255.255.255.255**)

In the DHCP **Request** message, the client uses an option called 'Option **54** - DHCP Server Identifier' to indicate which DHCP server it is communicating with.

4. DHCP Ack

The DHCP **Ack** message is sent by the DHCP server as a response to the DHCP **Request**, informing the client that it has accepted its request, and will provide it with the requested IP address. Some of the message properties are:

- It can be a **broadcast** or a **unicast** message.
- The **source MAC** is the server's MAC address
- The **destination MAC** is the client's MAC address, which the server learned from the **Discover** message.
- The **source IP** is the IP address of the server
- The **destination IP** is the IP address that's being offered to the client

How does a client tell the servers if it wants a unicast or broadcast response?

The client informs the servers if it wants a unicast or broadcast response through the "**Bootp flag**" that's contained in the DHCP **Discover** message.

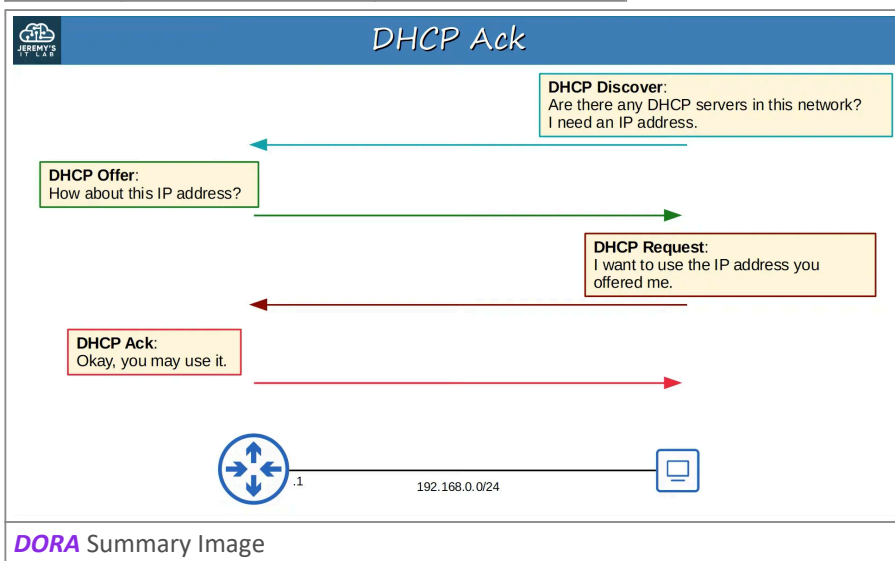
- If the **Bootp flag** value is **0x0000** then the response will be **unicast**.
- If the **Bootp flag** value is **0x8000** then the response will be **broadcast**.

Why would the client want a unicast or broadcast response?

Some clients may be configured in a way that does not let them accept **unicast** packets unless an IP address has already been assigned to them. In that case, the DHCP server is instructed to send the **Offer** message as a **broadcast**.

DHCP DORA + Release Table and Summary Image

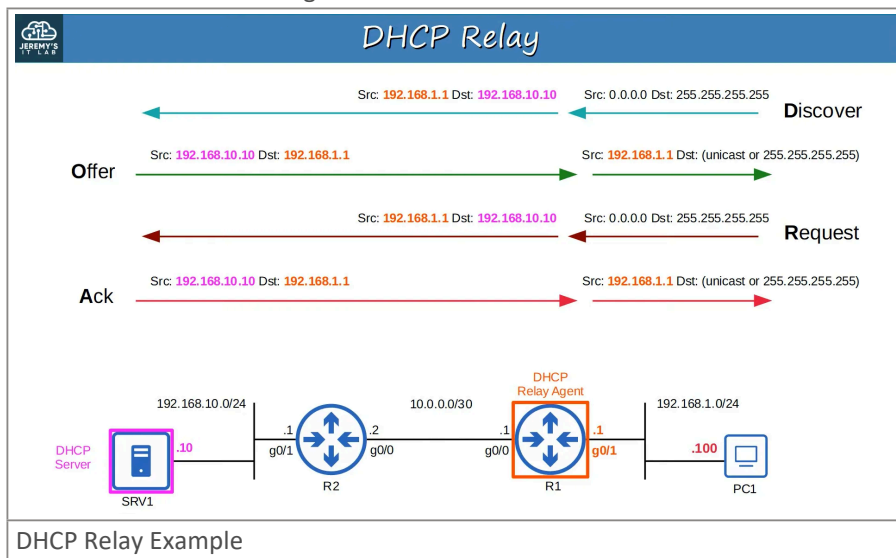
Step	Location --> Location	Message Type
Discover	Client --> Server	Broadcast
Offer	Server --> Client	Broadcast or Unicast
Request	Client --> Server	Broadcast
Ack	Server --> Client	Broadcast or Unicast
Release	Client --> Server	Unicast



DHCP Relay

Smaller networks may opt to use one of the router's as a DHCP server, but in bigger networks, a centralized DHCP server is often used. This is when DHCP **Relays** are used.

- Having a centralized **DHCP** server means that the server won't be able to receive the clients' **broadcast DHCP** messages. (**Broadcast** messages don't leave the local subnet)
- To fix this issue, we can configure routers to act as a **DHCP relay agent**.
 - A **relay agent** will forward the client's **broadcast DHCP** messages to the remote centralized **DHCP** server in the form of **unicast** messages.



DHCP and Windows OS

There are a few commands you can use on Windows to view DHCP information, as well as release and renew DHCP leases.

- **ipconfig /all**
 - Shows the IP configuration settings on Windows, including:
 - The **current IP address** and whether it's '**preferred**' or not (An IP address being **preferred** means that it has been assigned to the device by **DHCP** previously)
 - The **lease** obtain and expiration dates
 - The current **DHCP** server
- **ipconfig /release**
 - Sends a **DHCP Release** message to the **DHCP** server, informing it that it'd like to **release** the current IP address. The **DHCP** server will now consider this IP address to be **free** and usable by other devices.
 - Some of the **Release** message's properties are:
 - It is a **unicast** message.
 - The **source MAC** is the client's MAC address
 - The **destination MAC** is the server's MAC address
 - The **source IP** is the current IP address configured on the device
 - The **destination IP** is the IP address of the **DHCP** server
- **ipconfig /renew**
 - Initiates a communication with a **DHCP** server by following the **DORA** steps.
 - By the end of it, the client will have received an IP address from the server.

DHCP Server Configuration in Cisco IOS

DHCP Show Command

There's one show command to see the hosts which have been assigned an IP through the DHCP protocol

- **show ip dhcp binding**
 - Shows the hosts that have been assigned an IP address from the **DHCP** server
 - Displays information about the hosts including:

- *IP address*
- *Client ID / MAC address / Username*
- *Lease expiration date*

DHCP Server Configuration

DHCP server configuration is done through some commands in global config mode, and a few through DHCP-config mode.

1. Global Config

- *ip dhcp excluded-address LOW-ADDRESS HIGH-ADDRESS*
 - Specifies the range of addresses that won't be given to **DHCP** clients
- *ip dhcp pool POOL-NAME*
 - Initiates a **DHCP pool** and enters **DHCP pool** configuration mode or enters **DHCP pool** configuration if the **pool** already exists

2. DHCP Config

- *network IP-ADDRESS { /PREFIX-LENGTH | SUBNET-MASK }*
 - Specifies the subnet of addresses to be assigned to clients (Except the excluded addresses)
- *dns-server IP-ADDRESS*
 - Specifies the **DNS** server that **DHCP** clients should use
- *domain-name DOMAIN-NAME*
 - Specifies the *domain name* of the network
- *default-router IP-ADDRESS*
 - Specifies the *default gateway* that the **DHCP** clients should use
- *lease TIME { days hours minutes | infinite }*
 - Specifies the **DHCP** IP address *lease* time

DHCP Router Relay Agent Configuration

To configure a router to be a DHCP *relay agent*, you'd have to enter the configuration mode of an interface that's connected to clients and use the following command:

- *ip helper-address IP-ADDRESS*
 - Configures the current interface as a **DHCP relay agent**.
 - It will then forward **DHCP Discover** messages to the **DHCP** server in the form of *unicast* messages. (Instead of being *broadcast*)